

Idempotency Test

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A. Product of Vectors

$$C = A \wedge B, A \in \mathcal{B}(\mathcal{H}^1), B \in \mathcal{H}^1$$

$$\begin{aligned}
A &= \sum_{1 \leq j < k \leq 4} A_{jk} |\varphi_j \varphi_k\rangle \sum_{1 \leq i \leq 4} B_i |\varphi_i\rangle = \frac{1}{2} \sum_{1 \leq i, j, k \leq 4} A_{jk} B_i |\varphi_j \varphi_k \varphi_i\rangle \\
&= \frac{1}{2} \{ (A_{12}B_3 - A_{13}B_2 + A_{23}B_1 - A_{21}B_3 - A_{32}B_1 - A_{31}B_2) |123\rangle \\
&\quad + (A_{12}B_4 - A_{14}B_2 + A_{24}B_1 - A_{21}B_4 - A_{42}B_1 - A_{41}B_2) |124\rangle \\
&\quad + (A_{13}B_4 - A_{14}B_3 + A_{34}B_1 - A_{31}B_4 - A_{43}B_1 - A_{41}B_3) |134\rangle \\
&\quad + (A_{23}B_4 - A_{24}B_3 + A_{34}B_2 - A_{32}B_4 - A_{43}B_2 - A_{42}B_3) |234\rangle \}. \tag{1}
\end{aligned}$$

In more detail

$$\begin{aligned}
A &= \sum_{1 \leq j < k \leq 4} A_{jk} |\varphi_j \varphi_k\rangle \sum_{1 \leq i \leq 4} B_i |\varphi_i\rangle = \frac{1}{2} \sum_{1 \leq i, j, k \leq 4} A_{jk} B_i |\varphi_j \varphi_k \varphi_i\rangle \\
&= \frac{1}{2} (A_{12}B_3 |123\rangle + A_{13}B_2 |132\rangle + A_{23}B_1 |231\rangle) = (A_{12}B_3 - A_{13}B_2 + A_{23}B_1) |123\rangle \\
&\quad + \frac{1}{2} (A_{12}B_4 |124\rangle + A_{14}B_2 |142\rangle + A_{24}B_1 |241\rangle) = (A_{12}B_4 - A_{14}B_2 + A_{24}B_1) |124\rangle \\
&\quad + \frac{1}{2} (A_{13}B_4 |134\rangle + A_{14}B_3 |143\rangle + A_{34}B_1 |341\rangle) = (A_{13}B_4 - A_{14}B_3 + A_{34}B_1) |134\rangle \\
&\quad + \frac{1}{2} (A_{23}B_4 |234\rangle + A_{24}B_3 |243\rangle + A_{34}B_2 |342\rangle) = (A_{23}B_4 - A_{24}B_3 + A_{34}B_2) |234\rangle \tag{2}
\end{aligned}$$

and the coefficients of the unordered basis elements *are antisymmetric*.

B. Product of Operators

$$C = A \wedge B, A, B \in \mathcal{B}(\mathcal{H}^1)$$

$$A = \sum_{1 \leq j < k \leq r} A_{jk} |\varphi_j\rangle \langle \varphi_k| \sum_{1 \leq i < s \leq r} B_{is} |\varphi_i\rangle \langle \varphi_s| = \frac{1}{4} \sum_{\substack{1 \leq j, k \leq r \\ 1 \leq i, s \leq r}} A_{jk} B_{is} |\varphi_j \varphi_i\rangle \langle \varphi_k \varphi_s| \tag{3}$$

$$= \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq i < s \leq r}} \frac{1}{4} (A_{jk}B_{is} - A_{ik}B_{js} - A_{js}B_{ik} + A_{is}B_{jk}) |\varphi_j \varphi_i\rangle \langle \varphi_k \varphi_s| \quad (4)$$

$$C = A \wedge B, A \in \mathcal{B}(\mathcal{H}^2), B \in \mathcal{B}(\mathcal{H}^1)$$

$$A = \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} A_{jklm} |\varphi_j \varphi_k\rangle \langle \varphi_l \varphi_m| \sum_{1 \leq i, s \leq r} B_{is} |\varphi_i\rangle \langle \varphi_s| = \frac{1}{4} \sum_{\substack{1 \leq j, k \leq r \\ 1 \leq l, m \leq r \\ 1 \leq i, s \leq r}} A_{jklm} B_{is} |\varphi_j \varphi_k \varphi_i\rangle \langle \varphi_l \varphi_m \varphi_s| \quad (5)$$

all permutations of $\{jki\}$ and $\{lms\}$ are allowed for each $|\varphi_j \varphi_k \varphi_i\rangle \langle \varphi_l \varphi_m \varphi_s|$ this gives

$$\begin{aligned} &= \sum_{\substack{1 \leq j < k < i \leq r \\ 1 \leq l < m < s \leq r}} \frac{1}{4} (A_{jklm}B_{is} - A_{kjlm}B_{is} + A_{kilm}B_{js} - A_{iklm}B_{js} + A_{ijlm}B_{ks} - A_{jilm}B_{ks} \\ &\quad - A_{jkml}B_{is} + A_{kjml}B_{is} - A_{kiml}B_{js} + A_{ikml}B_{js} - A_{ijml}B_{ks} + A_{jiml}B_{ks} \\ &\quad + A_{jkms}B_{il} - A_{kjms}B_{il} + A_{kims}B_{jl} - A_{ikms}B_{jl} + A_{ijms}B_{kl} - A_{jims}B_{kl} \\ &\quad - A_{jksm}B_{il} + A_{kjsm}B_{il} - A_{kism}B_{jl} + A_{iksm}B_{jl} - A_{ijsm}B_{kl} + A_{jism}B_{kl} \\ &\quad + A_{jksl}B_{im} - A_{kjsl}B_{im} + A_{kisl}B_{jm} - A_{iksl}B_{jm} + A_{ijsl}B_{km} - A_{jisl}B_{km} \\ &\quad - A_{jkls}B_{im} + A_{kjls}B_{im} - A_{kils}B_{jm} + A_{ikls}B_{jm} - A_{ijls}B_{km} + A_{jils}B_{km}) |\varphi_j \varphi_k \varphi_i\rangle \langle \varphi_l \varphi_m \varphi_s| \end{aligned} \quad (6)$$

which gives

$$= \sum_{\substack{1 \leq j < k < i \leq r \\ 1 \leq l < m < s \leq r}} (A_{jklm}B_{is} + A_{kilm}B_{js} - A_{jilm}B_{ks} + A_{jkms}B_{il} + A_{kims}B_{jl} - A_{jims}B_{kl} \quad (7)$$

$$- A_{jkls}B_{im} - A_{kils}B_{jm} + A_{jils}B_{km}) |\varphi_j \varphi_k \varphi_i\rangle \langle \varphi_l \varphi_m \varphi_s|. \quad (8)$$

Thus

$$A \wedge B = \sum_{\substack{1 \leq j < k < i \leq r \\ 1 \leq l < m < s \leq r}} C_{jkilms} |\varphi_j \varphi_k \varphi_i\rangle \langle \varphi_l \varphi_m \varphi_s| \quad (9)$$

$$= \frac{1}{36} \sum_{\substack{1 \leq j, k, i \leq r \\ 1 \leq l, m, s \leq r}} C_{jkilms} |\varphi_j \varphi_k \varphi_i\rangle \langle \varphi_l \varphi_m \varphi_s|, \quad (10)$$

where

$$\begin{aligned} C_{jkilms} &= A_{jklm}B_{is} + A_{kilm}B_{js} - A_{jilm}B_{ks} + A_{jkms}B_{il} + A_{kims}B_{jl} - \\ &\quad A_{jims}B_{kl} - A_{jkls}B_{im} - A_{kils}B_{jm} + A_{jils}B_{km} \\ &\quad 1 \leq j, k, i \leq r, 1 \leq l, m, s \leq r. \end{aligned} \quad (11)$$

In particular this gives for $r = 4$

$$\begin{aligned}
A \wedge B = & (A_{1212}B_{33} - A_{1312}B_{23} + A_{2312}B_{13} - A_{1213}B_{32} + A_{1313}B_{22} - A_{2313}B_{12} \\
& + A_{1223}B_{31} - A_{1323}B_{21} + A_{2323}B_{11}) |123\rangle \langle 123| \\
& (A_{1212}B_{34} - A_{1312}B_{24} + A_{2312}B_{14} - A_{1214}B_{32} + A_{1314}B_{22} - A_{2314}B_{12} \\
& + A_{1224}B_{31} - A_{1324}B_{21} + A_{2324}B_{11}) |123\rangle \langle 124| \\
& (A_{1213}B_{34} - A_{1313}B_{24} + A_{2313}B_{14} - A_{1214}B_{33} + A_{1314}B_{23} - A_{2314}B_{13} \\
& + A_{1234}B_{31} - A_{1334}B_{21} + A_{2334}B_{11}) |123\rangle \langle 134| \\
& (A_{1223}B_{34} - A_{1323}B_{24} + A_{2323}B_{14} - A_{1224}B_{33} + A_{1324}B_{23} - A_{2324}B_{13} \\
& + A_{1234}B_{32} - A_{1334}B_{22} + A_{2334}B_{12}) |123\rangle \langle 234|
\end{aligned} \tag{12}$$

$$\begin{aligned}
+ & (A_{1212}B_{43} - A_{1412}B_{23} + A_{2412}B_{13} - A_{1214}B_{32} + A_{1414}B_{22} - A_{2313}B_{12} \\
& + A_{1223}B_{41} - A_{1423}B_{21} + A_{2423}B_{11}) |124\rangle \langle 123| \\
& (A_{1212}B_{44} - A_{1412}B_{24} + A_{2412}B_{14} - A_{1214}B_{42} + A_{1414}B_{22} - A_{2414}B_{12} \\
& + A_{1224}B_{41} - A_{1424}B_{21} + A_{2424}B_{11}) |124\rangle \langle 124| \\
& (A_{1213}B_{44} - A_{1413}B_{24} + A_{2413}B_{14} - A_{1214}B_{43} + A_{1414}B_{23} - A_{2414}B_{13} \\
& + A_{1234}B_{41} - A_{1434}B_{21} + A_{2434}B_{11}) |124\rangle \langle 134| \\
& (A_{1223}B_{44} - A_{1423}B_{24} + A_{2423}B_{14} - A_{1224}B_{43} + A_{1424}B_{23} - A_{2424}B_{13} \\
& + A_{1234}B_{42} - A_{1434}B_{22} + A_{2434}B_{12}) |124\rangle \langle 234|
\end{aligned} \tag{13}$$

$$\begin{aligned}
& (A_{1312}B_{43} - A_{1412}B_{33} + A_{3412}B_{13} - A_{1314}B_{32} + A_{1414}B_{32} - A_{2313}B_{12} \\
& + A_{1323}B_{41} - A_{1423}B_{31} + A_{3423}B_{11}) |134\rangle \langle 123| \\
& (A_{1312}B_{44} - A_{1412}B_{34} + A_{3412}B_{14} - A_{1314}B_{42} + A_{1414}B_{32} - A_{3414}B_{12} \\
& + A_{1324}B_{41} - A_{1424}B_{31} + A_{3424}B_{11}) |134\rangle \langle 124| \\
& (A_{1313}B_{44} - A_{1413}B_{34} + A_{3413}B_{14} - A_{1314}B_{43} + A_{1414}B_{33} - A_{3414}B_{13} \\
& + A_{1334}B_{41} - A_{1434}B_{31} + A_{3434}B_{11}) |134\rangle \langle 134| \\
& (A_{1323}B_{44} - A_{1423}B_{34} + A_{3423}B_{14} - A_{1324}B_{43} + A_{1424}B_{33} - A_{3424}B_{13} \\
& + A_{1334}B_{42} - A_{1434}B_{32} + A_{3434}B_{12}) |134\rangle \langle 234|
\end{aligned} \tag{14}$$

$$\begin{aligned}
& (A_{2312}B_{43} - A_{2412}B_{33} + A_{3412}B_{23} - A_{2314}B_{32} + A_{2414}B_{32} - A_{2313}B_{22} \\
& + A_{2323}B_{41} - A_{2423}B_{31} + A_{3423}B_{21}) |234\rangle \langle 123| \\
& (A_{2312}B_{44} - A_{2412}B_{34} + A_{3412}B_{24} - A_{2314}B_{42} + A_{2414}B_{32} - A_{3414}B_{22} \\
& + A_{2324}B_{41} - A_{2424}B_{31} + A_{3424}B_{21}) |234\rangle \langle 124| \\
& (A_{2313}B_{44} - A_{2413}B_{34} + A_{3413}B_{24} - A_{2314}B_{43} + A_{2414}B_{33} - A_{3414}B_{23} \\
& + A_{2334}B_{41} - A_{2434}B_{31} + A_{3434}B_{21}) |234\rangle \langle 134| \\
& (A_{2323}B_{44} - A_{2423}B_{34} + A_{3423}B_{24} - A_{2324}B_{43} + A_{2424}B_{33} - A_{3424}B_{23} \\
& + A_{2334}B_{42} - A_{2434}B_{32} + A_{3434}B_{22}) |234\rangle \langle 234|
\end{aligned} \tag{15}$$

If we let B be the projector onto $\mathcal{H}^1$ we obtain

$$\begin{aligned}
& = (A_{1212} + A_{1313} + A_{2323}) |123\rangle \langle 123| + (A_{1314} + A_{2324}) |123\rangle \langle 124| \\
& + (A_{2334} - A_{1214}) |123\rangle \langle 134| - (A_{1224} + A_{1334}) |123\rangle \langle 234|
\end{aligned} \tag{16}$$

$$\begin{aligned}
& + (A_{1414} + A_{2423}) |124\rangle \langle 123| + (A_{1212} + A_{1414} + A_{2424}) |124\rangle \langle 124| \\
& + (A_{1213} + A_{2434}) |124\rangle \langle 134| + (A_{1223} - A_{1434}) |124\rangle \langle 234|
\end{aligned} \tag{17}$$

$$\begin{aligned}
& + (A_{3423} - A_{1412}) |134\rangle \langle 123| + (A_{1312} + A_{3424}) |134\rangle \langle 124| \\
& + (A_{1313} + A_{1414} + A_{3434}) |134\rangle \langle 134| + (A_{1323} + A_{1424}) |134\rangle \langle 234|
\end{aligned} \tag{18}$$

$$\begin{aligned}
& - (A_{2412} + A_{2313}) |234\rangle \langle 123| + (A_{2312} - A_{3414}) |234\rangle \langle 124| \\
& + (A_{2313} + A_{2414}) |234\rangle \langle 134| + (A_{2323} + A_{2424} + A_{3434}) |234\rangle \langle 234|
\end{aligned} \tag{19}$$

which in general is given by

$$\begin{aligned}
C_{jkilms} &= A_{jklm}\delta_{is} + A_{kilm}\delta_{js} - A_{jilm}\delta_{ks} + A_{jkms}\delta_{il} + A_{kims}\delta_{jl} - \\
& A_{jims}\delta_{kl} - A_{jkls}\delta_{im} - A_{kils}\delta_{jm} + A_{jils}\delta_{km} \\
& 1 \leq j, k, i \leq r, 1 \leq l, m, s \leq r.
\end{aligned} \tag{20}$$

We will need to consider

$$(A \wedge B)^2 = \sum_{\substack{1 \leq j < k < i \leq r \\ 1 \leq l < m < s \leq r \\ 1 \leq l' < m' < s' \leq r}} C_{jkilms} C_{lmsl'm's'} |\varphi_j \varphi_k \varphi_i\rangle \langle \varphi_{l'} \varphi_{m'} \varphi_{s'}|. \tag{21}$$

$$P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} = \sum_{1 \leq i_1, i_2, i_3 \leq r} |\varphi_{i_1}\rangle \langle \varphi_{i_1}| \wedge |\varphi_{i_2}\rangle \langle \varphi_{i_2}| \wedge |\varphi_{i_3}\rangle \langle \varphi_{i_3}| \quad (22)$$

$$= \sum_{1 \leq i_1, i_2, i_3 \leq r} |\varphi_{i_1} \varphi_{i_2} \varphi_{i_3}\rangle \langle \varphi_{i_1} \varphi_{i_2} \varphi_{i_3}| = 6P_{\mathcal{H}^3} \quad (23)$$

$$P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} = 2P_{\mathcal{H}^2} \quad (24)$$

The general condition for idempotency of a particle conserving operator is

$$P_{\mathcal{H}^m} A P_{\mathcal{H}^m} = P_{\mathcal{H}^m} A P_{\mathcal{H}^m} A P_{\mathcal{H}^m}; 0 \leq m \leq r \quad (25)$$

The action of the projector $P_{\mathcal{H}^3}$ on A is given by

$$\begin{aligned} & |\varphi_{j_1} \varphi_{j_2} \varphi_{j_3}\rangle \langle \varphi_{j_1} \varphi_{j_2} \varphi_{j_3}| A |\varphi_{k_1} \varphi_{k_2} \varphi_{k_3}\rangle \langle \varphi_{k_1} \varphi_{k_2} \varphi_{k_3}| \\ &= |\varphi_{j_1} \varphi_{j_2} \varphi_{j_3}\rangle \langle \varphi_{j_1} \varphi_{j_2} \varphi_{j_3}| A |\varphi_{k_1} \varphi_{k_2} \varphi_{k_3}\rangle \langle \varphi_{k_1} \varphi_{k_2} \varphi_{k_3}| A |\varphi_{l_1} \varphi_{l_2} \varphi_{l_3}\rangle \langle \varphi_{l_1} \varphi_{l_2} \varphi_{l_3}|. \end{aligned} \quad (26)$$

If A is the second quantization of a two particle operator then

$$A = \Gamma(A_2) = \Gamma\left(\sum_{p_1, p_2, q_1, q_2} A_2 |\varphi_{p_1} \varphi_{p_2}\rangle \langle \varphi_{q_1} \varphi_{q_2}| \right) = A_2 \wedge I_{\mathcal{H}_{\mathcal{F}}}$$

and

$$P_{\mathcal{H}^2} A P_{\mathcal{H}^2} = P_{\mathcal{H}^2} \Gamma(A_2) P_{\mathcal{H}^2} = \frac{1}{4} \sum_{1 \leq p_1, p_2, q_1, q_2 \leq r} A_{2p_1 p_2 q_1 q_2} |\varphi_{p_1} \varphi_{p_2}\rangle \langle \varphi_{q_1} \varphi_{q_2}|. \quad (27)$$

An important relationship is given by

$$\begin{aligned} P_{\mathcal{H}^3} A P_{\mathcal{H}^3} &= P_{\mathcal{H}^3} (A_2 \wedge I_{\mathcal{H}_{\mathcal{F}}}) P_{\mathcal{H}^3} = A_2 \wedge P_{\mathcal{H}^1} \\ &= \frac{1}{4} \sum_{1 \leq p_1, p_2, q_1, q_2, i \leq r} A_{2p_1 p_2 q_1 q_2} |\varphi_{p_1} \varphi_{p_2}\rangle \langle \varphi_{q_1} \varphi_{q_2}| \wedge |\varphi_i\rangle \langle \varphi_i|. \end{aligned} \quad (28)$$

The squared operator is

$$P_{\mathcal{H}^3} A P_{\mathcal{H}^3} A P_{\mathcal{H}^3} = \frac{1}{16} \sum_{\substack{1 \leq p_1, p_2, q_1, q_2, i \leq r \\ 1 \leq p'_1, p'_2, q'_1, q'_2, i' \leq r}} A_{2p_1 p_2 q_1 q_2} A_{2p'_1 p'_2 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q_1} \varphi_{q_2} \varphi_i| |\varphi_{p'_1} \varphi_{p'_2} \varphi_{i'}\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_{i'}| \quad (29)$$

as

$$\begin{aligned} \langle \varphi_{q_1} \varphi_{q_2} \varphi_i | \varphi_{p'_1} \varphi_{p'_2} \varphi_{i'} \rangle &= \langle \varphi_{q_1} \varphi_{q_2} \varphi_i | \mathcal{A}_3 \varphi_{p'_1} \varphi_{p'_2} \varphi_{i'} \rangle \\ &= \frac{1}{6} \{ \delta_{q_1 p'_1} \delta_{q_2 p'_2} \delta_{ii'} - \delta_{q_1 p'_2} \delta_{q_2 p'_1} \delta_{ii'} - \delta_{q_1 p'_1} \delta_{q_2 i'} \delta_{i p'_2} \\ &\quad + \delta_{q_1 i'} \delta_{q_2 p'_1} \delta_{i p'_2} + \delta_{q_1 p'_2} \delta_{q_2 i'} \delta_{i p'_1} - \delta_{q_1 i'} \delta_{q_2 p'_2} \delta_{i p'_1} \} \end{aligned} \quad (30)$$

[

$$|\varphi_1\varphi_2\varphi_3\rangle = \frac{1}{\sqrt{6}} (\varphi_1 \otimes \varphi_2 \otimes \varphi_3 - \varphi_2 \otimes \varphi_1 \otimes \varphi_3 + \varphi_2 \otimes \varphi_3 \otimes \varphi_1 - \varphi_3 \otimes \varphi_2 \otimes \varphi_1$$
(31)

$$+ \varphi_3 \otimes \varphi_1 \otimes \varphi_2 - \varphi_1 \otimes \varphi_3 \otimes \varphi_2)$$
(32)

$$\langle \varphi_1\varphi_2\varphi_3 | \varphi_1\varphi_2\varphi_3 \rangle = 1$$
(33)

] one has

$$\begin{aligned} & P_{\mathcal{H}^3} A P_{\mathcal{H}^3} A P_{\mathcal{H}^3} \\ &= \frac{1}{96} \sum_{\substack{1 \leq p_1, p_2, q_1, q_2, i \leq r \\ 1 \leq p'_1, p'_2, q'_1, q'_2, i' \leq r}} A_{2p_1 p_2 q_1 q_2} A_{2p'_1 p'_2 q'_1 q'_2} \{ \delta_{q_1 p'_1} \delta_{q_2 p'_2} \delta_{ii'} - \delta_{q_1 p'_2} \delta_{q_2 p'_1} \delta_{ii'} - \delta_{q_1 p'_1} \delta_{q_2 i'} \delta_{ip'_2} \\ &+ \delta_{q_1 i'} \delta_{q_2 p'_1} \delta_{ip'_2} + \delta_{q_1 p'_2} \delta_{q_2 i'} \delta_{ip'_1} - \delta_{q_1 i'} \delta_{q_2 p'_2} \delta_{ip'_1} \} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_{i'}| \end{aligned}$$
(34)

which gives

$$\begin{aligned} P_{\mathcal{H}^3} A P_{\mathcal{H}^3} A P_{\mathcal{H}^3} &= \frac{1}{96} \left\{ \sum_{\substack{p_1, p_2, q_1, q_2, i, \\ q'_1, q'_2}} A_{2p_1 p_2 q_1 q_2} A_{2q_1 q_2 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_i| \right. \\ &- \sum_{\substack{p_1, p_2, q_1, q_2, i, \\ q'_1, q'_2}} A_{2p_1 p_2 q_1 q_2} A_{2q_2 q_1 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_i| \end{aligned}$$
(35)

$$- \sum_{\substack{p_1, p_2, q_1, i, \\ q'_1, q'_2, i'}} A_{2p_1 p_2 i' q_1} A_{2iq_1 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_{i'}|$$
(36)

$$- \sum_{\substack{p_1, p_2, q_1, i, \\ q'_1, q'_2, i'}} A_{2p_1 p_2 i' q_1} A_{2iq_1 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_{i'}|$$
(37)

$$- \sum_{\substack{p_1, p_2, q_1, i, \\ q'_1, q'_2, i'}} A_{2p_1 p_2 i' q_1} A_{2iq_1 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_{i'}|$$
(38)

$$- \sum_{\substack{p_1, p_2, q_1, i, \\ q'_1, q'_2, i'}} A_{2p_1 p_2 i' q_1} A_{2iq_1 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_{i'}| \Bigg\}$$
(39)

$$P_{\mathcal{H}^3} A P_{\mathcal{H}^3} A P_{\mathcal{H}^3} = \frac{1}{48} \sum_{\substack{1 \leq p_1, p_2, i, \\ q'_1, q'_2 \leq 3}} A_{2p_1 p_2 q_1 q_2} A_{2q_1 q_2 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_i| \quad (40)$$

$$- \frac{1}{24} \sum_{\substack{1 \leq p_1, p_2, q_1, i, \\ q'_1, q'_2, i' \leq 3}} A_{2p_1 p_2 i' q_1} A_{2i q_1 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_{i'}| \quad (41)$$

which gives

$$P_{\mathcal{H}^3} A P_{\mathcal{H}^3} A P_{\mathcal{H}^3} = \frac{1}{48} \sum_{\substack{1 \leq p_1, p_2, i, \\ q'_1, q'_2 \leq 3}} A_{2p_1 p_2 q'_1 q'_2}^2 |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_i| \quad (42)$$

$$- \frac{1}{24} \sum_{\substack{1 \leq p_1, p_2, q_1, i, \\ q'_1, q'_2, i' \leq 3}} A_{2p_1 p_2 i' q_1} A_{2i q_1 q'_1 q'_2} |\varphi_{p_1} \varphi_{p_2} \varphi_i\rangle \langle \varphi_{q'_1} \varphi_{q'_2} \varphi_{i'}| \quad (43)$$

$$\begin{aligned} (A \wedge B)^2 &= \sum_{\substack{1 \leq j < k < i \leq r \\ 1 \leq l < m < s \leq r \\ 1 \leq l' < m' < s' \leq r}} C_{jkilms} C_{lmsl'm's'} |\varphi_j \varphi_k \varphi_i\rangle \langle \varphi_{l'} \varphi_{m'} \varphi_{s'}| \\ &= \sum_{\substack{1 \leq j < k < i \leq r \\ 1 \leq l' < m' < s' \leq r}} (C^2)_{jkil'm's'} |\varphi_j \varphi_k \varphi_i\rangle \langle \varphi_{l'} \varphi_{m'} \varphi_{s'}|. \end{aligned} \quad (44)$$

where

$$\begin{aligned} C_{jkilms} &= A_{jklm} B_{is} + A_{kilm} B_{js} - A_{jilm} B_{ks} + A_{jkms} B_{il} + A_{kims} B_{jl} - \\ &\quad A_{jims} B_{kl} - A_{jkls} B_{im} - A_{kils} B_{jm} + A_{jils} B_{km} \\ &\quad 1 \leq j, k, i \leq r, 1 \leq l, m, s \leq r. \end{aligned} \quad (45)$$

and

$$(C^2)_{jkil'm's'} = \sum_{1 \leq l < m < s \leq r} C_{jkilms} C_{lmsl'm's'} = \frac{1}{36} \sum_{1 \leq l, m, s \leq r} C_{jkilms} C_{lmsl'm's'}. \quad (46)$$

In the case $B = P_{\mathcal{H}^2}$ one has

$$\begin{aligned} C_{jkilms} &= A_{jklm} \delta_{is} + A_{kilm} \delta_{js} - A_{jilm} \delta_{ks} + A_{jkms} \delta_{il} + A_{kims} \delta_{jl} - \\ &\quad A_{jims} \delta_{kl} - A_{jkls} \delta_{im} - A_{kils} \delta_{jm} + A_{jils} \delta_{km} \\ &\quad 1 \leq j, k, i \leq r, 1 \leq l, m, s \leq r. \end{aligned} \quad (47)$$

$$\begin{aligned}
C_{lmsl'm's'} &= A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'} + A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - \\
&\quad A_{lsm's'}\delta_{ml'} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'} \\
1 \leq j, k, i \leq r, 1 \leq l, m, s \leq r.
\end{aligned} \tag{48}$$

thus

$$(C^2)_{jkil'm's'} = \frac{1}{36} \sum_{1 \leq l, m, s \leq r} (A_{jklm}\delta_{is} + A_{kilm}\delta_{js} - A_{jilm}\delta_{ks} + A_{jkms}\delta_{il} + A_{kims}\delta_{jl} - \tag{49}$$

$$A_{jims}\delta_{kl} - A_{jkls}\delta_{im} - A_{kils}\delta_{jm} + A_{jils}\delta_{km}) (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'} \tag{50}$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml'} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \tag{51}$$

1

$$(C^2)_{jkil'm's'} = \frac{1}{36} \sum_{1 \leq l, m, s \leq r} A_{jklm}\delta_{is} (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'} \tag{52}$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml'} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \tag{53}$$

2

$$+ A_{kilm}\delta_{js} (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'} \tag{54}$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml'} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \tag{55}$$

3

$$- A_{jilm}\delta_{ks} (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'} \tag{56}$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml'} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \tag{57}$$

4

$$+ A_{jkms}\delta_{il} (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'} \tag{58}$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml'} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \tag{59}$$

5

$$- A_{kims}\delta_{jl} (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'} \tag{60}$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml'} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \tag{61}$$

6

$$A_{jims}\delta_{kl} (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'}) \quad (62)$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \quad (63)$$

7

$$- A_{jkl}\delta_{im} (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'}) \quad (64)$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \quad (65)$$

8

$$- A_{kils}\delta_{jm} (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'}) \quad (66)$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \quad (67)$$

9

$$+ A_{jils}\delta_{km} (A_{lml'm'}\delta_{ss'} + A_{msl'm'}\delta_{ls'} - A_{lsl'm'}\delta_{ms'}) \quad (68)$$

$$+ A_{lmm's'}\delta_{sl'} + A_{msm's'}\delta_{ll'} - A_{lsm's'}\delta_{ml} - A_{lml's'}\delta_{sm'} - A_{msl's'}\delta_{lm'} + A_{lsl's'}\delta_{mm'}) \quad (69)$$

1

$$(C^2)_{jkil'm's'} = \frac{1}{36} \sum_{1 \leq l, m, s \leq r} (A_{jklm}A_{lml'm'}\delta_{is}\delta_{ss'} + A_{jklm}A_{msl'm'}\delta_{is}\delta_{ls'} - A_{jklm}A_{lsl'm'}\delta_{is}\delta_{ms'} \quad (70)$$

$$+ A_{jklm}A_{lmm's'}\delta_{is}\delta_{sl'} + A_{jklm}A_{msm's'}\delta_{is}\delta_{ll'} - A_{jklm}A_{lsm's'}\delta_{is}\delta_{ml} - A_{jklm}A_{lml's'}\delta_{is}\delta_{sm'} - A_{jklm}A_{msl's'}\delta_{is}\delta_{lm'} + A_{jklm}A_{lsl's'}\delta_{is}\delta_{mm'}) \quad (71)$$

2

$$+ (A_{kilm}A_{lml'm'}\delta_{js}\delta_{ss'} + A_{kilm}A_{msl'm'}\delta_{js}\delta_{ls'} - A_{kilm}A_{lsl'm'}\delta_{js}\delta_{ms'} \quad (72)$$

$$+ A_{kilm}A_{lmm's'}\delta_{js}\delta_{sl'} + A_{kilm}A_{msm's'}\delta_{js}\delta_{ll'} - A_{kilm}A_{lsm's'}\delta_{js}\delta_{ml} - A_{kilm}A_{lml's'}\delta_{js}\delta_{sm'} - A_{kilm}A_{msl's'}\delta_{js}\delta_{lm'} + A_{kilm}A_{lsl's'}\delta_{js}\delta_{mm'}) \quad (73)$$

3

$$- (A_{jilm}A_{lml'm'}\delta_{ks}\delta_{ss'} + A_{jilm}A_{msl'm'}\delta_{ks}\delta_{ls'} - A_{jilm}A_{lsl'm'}\delta_{ks}\delta_{ms'} \quad (74)$$

$$+ A_{jilm}A_{lmm's'}\delta_{ks}\delta_{sl'} + A_{jilm}A_{msm's'}\delta_{ks}\delta_{ll'} - A_{jilm}A_{lsm's'}\delta_{ks}\delta_{ml} - A_{jilm}A_{lml's'}\delta_{ks}\delta_{sm'} \\ - A_{jilm}A_{msl's'}\delta_{ks}\delta_{lm'} + A_{jilm}A_{ls's'}\delta_{ks}\delta_{mm'} \quad (75)$$

4

$$+ (A_{jkms}A_{lml'm'}\delta_{il}\delta_{ss'} + A_{jkms}A_{msl'm'}\delta_{il}\delta_{ls'} - A_{jkms}A_{lsl'm'}\delta_{il}\delta_{ms'} \quad (76)$$

$$+ A_{jkms}A_{lmm's'}\delta_{il}\delta_{sl'} + A_{jkms}A_{msm's'}\delta_{il}\delta_{ll'} - A_{jkms}A_{lsm's'}\delta_{il}\delta_{ml} - A_{jkms}A_{lml's'}\delta_{il}\delta_{sm'} \\ - A_{jkms}A_{msl's'}\delta_{il}\delta_{lm'} + A_{jkms}A_{ls's'}\delta_{il}\delta_{mm'} \quad (77)$$

5

$$+ (A_{kims}A_{lml'm'}\delta_{jl}\delta_{ss'} + A_{kims}A_{msl'm'}\delta_{jl}\delta_{ls'} - A_{kims}A_{lsl'm'}\delta_{jl}\delta_{ms'} \quad (78)$$

$$+ A_{kims}A_{lmm's'}\delta_{jl}\delta_{sl'} + A_{kims}A_{msm's'}\delta_{jl}\delta_{ll'} - A_{kims}A_{lsm's'}\delta_{jl}\delta_{ml} - A_{kims}A_{lml's'}\delta_{jl}\delta_{sm'} \\ - A_{kims}A_{msl's'}\delta_{jl}\delta_{lm'} + A_{kims}A_{ls's'}\delta_{jl}\delta_{mm'} \quad (79)$$

6

$$- (A_{jims}A_{lml'm'}\delta_{kl}\delta_{ss'} + A_{jims}A_{msl'm'}\delta_{kl}\delta_{ls'} - A_{jims}A_{lsl'm'}\delta_{kl}\delta_{ms'} \quad (80)$$

$$+ A_{jims}A_{lmm's'}\delta_{kl}\delta_{sl'} + A_{jims}A_{msm's'}\delta_{kl}\delta_{ll'} - A_{jims}A_{lsm's'}\delta_{kl}\delta_{ml} - A_{jims}A_{lml's'}\delta_{kl}\delta_{sm'} \\ - A_{jims}A_{msl's'}\delta_{kl}\delta_{lm'} + A_{jims}A_{ls's'}\delta_{kl}\delta_{mm'} \quad (81)$$

7

$$- (A_{jkls}A_{lml'm'}\delta_{im}\delta_{ss'} + A_{jkls}A_{msl'm'}\delta_{im}\delta_{ls'} - A_{jkls}A_{lsl'm'}\delta_{im}\delta_{ms'} \quad (82)$$

$$+ A_{jkls}A_{lmm's'}\delta_{im}\delta_{sl'} + A_{jkls}A_{msm's'}\delta_{im}\delta_{ll'} - A_{jkls}A_{lsm's'}\delta_{im}\delta_{ml} - A_{jkls}A_{lml's'}\delta_{im}\delta_{sm'} \\ - A_{jkls}A_{msl's'}\delta_{im}\delta_{lm'} + A_{jkls}A_{ls's'}\delta_{im}\delta_{mm'} \quad (83)$$

8

$$\begin{aligned}
& - (A_{kils}A_{lm'l'm'}\delta_{jm}\delta_{ss'} + A_{kils}A_{msl'm'}\delta_{jm}\delta_{ls'} - A_{kils}A_{lsl'm'}\delta_{jm}\delta_{ms'} \\
& + A_{kils}A_{lmm's'}\delta_{jm}\delta_{sl'} + A_{kils}A_{msm's'}\delta_{jm}\delta_{ll'} - A_{kils}A_{lsm's'}\delta_{jm}\delta_{ml} - A_{kils}A_{lm'l's'}\delta_{jm}\delta_{sm'} \\
& - A_{kils}A_{msl's'}\delta_{jm}\delta_{lm'} + A_{kils}A_{lsl's'}\delta_{jm}\delta_{mm'}) \quad (84)
\end{aligned}$$

9

$$\begin{aligned}
& + (A_{jils}A_{lm'l'm'}\delta_{km}\delta_{ss'} + A_{jils}A_{msl'm'}\delta_{km}\delta_{ls'} - A_{jils}A_{lsl'm'}\delta_{km}\delta_{ms'} \\
& + A_{jils}A_{lmm's'}\delta_{km}\delta_{sl'} + A_{jils}A_{msm's'}\delta_{km}\delta_{ll'} - A_{jils}A_{lsm's'}\delta_{km}\delta_{ml} - A_{jils}A_{lm'l's'}\delta_{km}\delta_{sm'} \\
& - A_{jils}A_{msl's'}\delta_{km}\delta_{lm'} + A_{jils}A_{lsl's'}\delta_{km}\delta_{mm'}) \quad (86)
\end{aligned}$$

1

$$(C^2)_{jkil'm's'} = \frac{1}{36} \sum_{1 \leq l, m, s \leq r} (A_{jkl'm'}^2 \delta_{is'} + A_{jks'm} A_{mil'm'} - A_{jklm} A_{lil's'}) \quad (88)$$

$$\begin{aligned}
& + A_{jkm'l'}^2 \delta_{is} + A_{jkl'm} A_{mim's'} + A_{jkl'l'} A_{ilm's'} - A_{jklml's'}^2 \delta_{im'} \\
& - A_{jkm'm} A_{mil's'} + A_{jklm'} A_{lil's'}) \quad (89)
\end{aligned}$$

2

$$\begin{aligned}
& + (A_{kilm}A_{lm'l'm'}\delta_{js}\delta_{ss'} + A_{kilm}A_{msl'm'}\delta_{js}\delta_{ls'} - A_{kilm}A_{lsl'm'}\delta_{js}\delta_{ms'} \\
& + A_{kilm}A_{lmm's'}\delta_{js}\delta_{sl'} + A_{kilm}A_{msm's'}\delta_{js}\delta_{ll'} - A_{kilm}A_{lsm's'}\delta_{js}\delta_{ml} - A_{kilm}A_{lm'l's'}\delta_{js}\delta_{sm'} \\
& - A_{kilm}A_{msl's'}\delta_{js}\delta_{lm'} + A_{kilm}A_{lsl's'}\delta_{js}\delta_{mm'}) \quad (90)
\end{aligned}$$

3

$$\begin{aligned}
& - (A_{jilm}A_{lm'l'm'}\delta_{ks}\delta_{ss'} + A_{jilm}A_{msl'm'}\delta_{ks}\delta_{ls'} - A_{jilm}A_{lsl'm'}\delta_{ks}\delta_{ms'} \\
& + A_{jilm}A_{lmm's'}\delta_{ks}\delta_{sl'} + A_{jilm}A_{msm's'}\delta_{ks}\delta_{ll'} - A_{jilm}A_{lsm's'}\delta_{ks}\delta_{ml} - A_{jilm}A_{lm'l's'}\delta_{ks}\delta_{sm'} \\
& - A_{jilm}A_{msl's'}\delta_{ks}\delta_{lm'} + A_{jilm}A_{lsl's'}\delta_{ks}\delta_{mm'}) \quad (92)
\end{aligned}$$

(93)

4

$$+ (A_{jkms}A_{lml'm'}\delta_{il}\delta_{ss'} + A_{jkms}A_{msl'm'}\delta_{il}\delta_{ls'} - A_{jkms}A_{lsl'm'}\delta_{il}\delta_{ms'} \quad (94)$$

$$+ A_{jkms}A_{lmm's'}\delta_{il}\delta_{sl'} + A_{jkms}A_{msm's'}\delta_{il}\delta_{ll'} - A_{jkms}A_{lsm's'}\delta_{il}\delta_{ml} - A_{jkms}A_{lml's'}\delta_{il}\delta_{sm'} \\ - A_{jkms}A_{msl's'}\delta_{il}\delta_{lm'} + A_{jkms}A_{ls's'}\delta_{il}\delta_{mm'} \quad (95)$$

5

$$+ (A_{kims}A_{lml'm'}\delta_{jl}\delta_{ss'} + A_{kims}A_{msl'm'}\delta_{jl}\delta_{ls'} - A_{kims}A_{lsl'm'}\delta_{jl}\delta_{ms'} \quad (96)$$

$$+ A_{kims}A_{lmm's'}\delta_{jl}\delta_{sl'} + A_{kims}A_{msm's'}\delta_{jl}\delta_{ll'} - A_{kims}A_{lsm's'}\delta_{jl}\delta_{ml} - A_{kims}A_{lml's'}\delta_{jl}\delta_{sm'} \\ - A_{kims}A_{msl's'}\delta_{jl}\delta_{lm'} + A_{kims}A_{ls's'}\delta_{jl}\delta_{mm'} \quad (97)$$

6

$$- (A_{jims}A_{lml'm'}\delta_{kl}\delta_{ss'} + A_{jims}A_{msl'm'}\delta_{kl}\delta_{ls'} - A_{jims}A_{lsl'm'}\delta_{kl}\delta_{ms'} \quad (98)$$

$$+ A_{jims}A_{lmm's'}\delta_{kl}\delta_{sl'} + A_{jims}A_{msm's'}\delta_{kl}\delta_{ll'} - A_{jims}A_{lsm's'}\delta_{kl}\delta_{ml} - A_{jims}A_{lml's'}\delta_{kl}\delta_{sm'} \\ - A_{jims}A_{msl's'}\delta_{kl}\delta_{lm'} + A_{jims}A_{ls's'}\delta_{kl}\delta_{mm'} \quad (99)$$

7

$$- (A_{jkls}A_{lml'm'}\delta_{im}\delta_{ss'} + A_{jkls}A_{msl'm'}\delta_{im}\delta_{ls'} - A_{jkls}A_{lsl'm'}\delta_{im}\delta_{ms'} \quad (100)$$

$$+ A_{jkls}A_{lmm's'}\delta_{im}\delta_{sl'} + A_{jkls}A_{msm's'}\delta_{im}\delta_{ll'} - A_{jkls}A_{lsm's'}\delta_{im}\delta_{ml} - A_{jkls}A_{lml's'}\delta_{im}\delta_{sm'} \\ - A_{jkls}A_{msl's'}\delta_{im}\delta_{lm'} + A_{jkls}A_{ls's'}\delta_{im}\delta_{mm'} \quad (101)$$

8

$$- (A_{kils}A_{lml'm'}\delta_{jm}\delta_{ss'} + A_{kils}A_{msl'm'}\delta_{jm}\delta_{ls'} - A_{kils}A_{lsl'm'}\delta_{jm}\delta_{ms'} \quad (102)$$

$$+ A_{kils}A_{lmm's'}\delta_{jm}\delta_{sl'} + A_{kils}A_{msm's'}\delta_{jm}\delta_{ll'} - A_{kils}A_{lsm's'}\delta_{jm}\delta_{ml} - A_{kils}A_{lml's'}\delta_{jm}\delta_{sm'} \\ - A_{kils}A_{msl's'}\delta_{jm}\delta_{lm'} + A_{kils}A_{ls's'}\delta_{jm}\delta_{mm'} \quad (103)$$

$$+ (A_{jils}A_{lm'l'm'}\delta_{km}\delta_{ss'} + A_{jils}A_{msl'm'}\delta_{km}\delta_{ls'} - A_{jils}A_{lsl'm'}\delta_{km}\delta_{ms'} \quad (104)$$

$$\left. \begin{aligned} &+ A_{jils}A_{lmm's'}\delta_{km}\delta_{sl'} + A_{jils}A_{msm's'}\delta_{km}\delta_{ll'} - A_{jils}A_{lsm's'}\delta_{km}\delta_{ml} - A_{jils}A_{lm'l's'}\delta_{km}\delta_{sm'} \\ &- A_{jils}A_{msl's'}\delta_{km}\delta_{lm'} + A_{jils}A_{lsl's'}\delta_{km}\delta_{mm'} \end{aligned} \right) \quad (105)$$

$$A = A^2 \quad (106)$$

$$\sum_{1 \leq j \leq r} A_{ij}A_{jk} = A_{ik} \quad (107)$$

$$\sum_{1 \leq j \leq r} A_{ij}A_{ji} = \sum_{1 \leq j \leq r} |A_{ij}|^2 = A_{ii} \geq 0 \quad (108)$$

$$\sum_{1 \leq j \neq i \leq r} |A_{ij}|^2 + (A_{ii})^2 = A_{ii} \quad (109)$$

$$\sum_{1 \leq j \neq i \leq r} |A_{ij}|^2 = A_{ii} - (A_{ii})^2 = A_{ii}(1 - A_{ii}) \quad (110)$$

$$\Rightarrow 1 \geq A_{ii} \geq 0 \quad (111)$$

$$A_2 = \sum_{1 \leq J \leq \binom{r}{2}} |g_J\rangle \langle g_J| = \frac{1}{4} \sum_{1 \leq J \leq \binom{r}{2}} \sum_{\substack{1 \leq j,k \leq r \\ 1 \leq l,m \leq r}} g_{Jjk}g_{Jlm} |\varphi_j\varphi_k\rangle \langle \varphi_l\varphi_m| \quad (112)$$

$$A_2 \wedge P_{\mathcal{H}^1} = \frac{1}{4} \sum_{1 \leq J \leq \binom{r}{2}} \sum_{1 \leq i,j,k,l,m \leq r} g_{Jjk}\bar{g}_{Jlm} |\varphi_i\varphi_j\varphi_k\rangle \langle \varphi_i\varphi_l\varphi_m| \quad (113)$$

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J,K \leq \binom{r}{2}} \sum_{\substack{1 \leq i,j,k,l,m \leq r \\ 1 \leq \alpha,\beta,\gamma,\zeta,\eta \leq r}} g_{Jjk}\bar{g}_{Jlm}g_{K\beta\gamma}\bar{g}_{K\zeta\eta} |\varphi_i\varphi_j\varphi_k\rangle \langle \varphi_i\varphi_l\varphi_m | \varphi_\alpha\varphi_\beta\varphi_\gamma\rangle \langle \varphi_\alpha\varphi_\zeta\varphi_\eta| \quad (114)$$

$$\begin{aligned} \langle \varphi_i\varphi_l\varphi_m | \varphi_\alpha\varphi_\beta\varphi_\gamma\rangle &= \frac{1}{6} \{ \delta_{i\alpha}\delta_{l\beta}\delta_{m\gamma} - \delta_{i\beta}\delta_{l\alpha}\delta_{m\gamma} - \delta_{i\alpha}\delta_{l\gamma}\delta_{m\beta} \\ &+ \delta_{i\gamma}\delta_{l\alpha}\delta_{m\beta} + \delta_{i\beta}\delta_{l\gamma}\delta_{m\alpha} - \delta_{i\gamma}\delta_{l\beta}\delta_{m\alpha} \} \end{aligned} \quad (115)$$

1

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (116)$$

$$\times \frac{1}{6} \{ g_{Jjk} \bar{g}_{Jlm} g_{Klm} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{Klm} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jlm} g_{Kml} \bar{g}_{K\zeta\eta} \delta_{i\alpha} \quad (117)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{Kmi} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} \bar{g}_{Kil} g_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{Kli} \bar{g}_{K\zeta\eta} \} \quad (118)$$

2

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (119)$$

$$\times \frac{1}{6} \{ g_{Jjk} \bar{g}_{Jlm} g_{Klm} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{Klm} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jlm} g_{Kml} \bar{g}_{K\zeta\eta} \delta_{i\alpha} \quad (120)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{Kmi} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} g_{Kil} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{Kli} \bar{g}_{K\zeta\eta} \} \quad (121)$$

3

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (122)$$

$$\times \frac{1}{6} \{ g_{Jjk} \bar{g}_{Jlm} g_{Klm} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{Klm} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jlm} g_{Kml} \bar{g}_{K\zeta\eta} \delta_{i\alpha} \quad (123)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{Kmi} \bar{g}_{K\zeta\eta} \delta_{i\alpha} + g_{Jjk} \bar{g}_{Jl\alpha} g_{Kil} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{Kli} \bar{g}_{K\zeta\eta} \} \quad (124)$$

4

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (125)$$

$$\times \frac{1}{6} \{ g_{Jjk} \bar{g}_{Jlm} g_{Klm} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{Klm} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jlm} g_{Kml} \bar{g}_{K\zeta\eta} \delta_{i\alpha} \quad (126)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{Kmi} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} g_{Kil} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{Kli} \bar{g}_{K\zeta\eta} \} \quad (127)$$

5

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (128)$$

$$\times \frac{1}{6} \{ g_{Jjk} \bar{g}_{Jlm} g_{Klm} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{Klm} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jlm} g_{Kml} \bar{g}_{K\zeta\eta} \delta_{i\alpha} \quad (129)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{Kmi} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} g_{Kil} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{Kli} \bar{g}_{K\zeta\eta} \} \quad (130)$$

6

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (131)$$

$$\times \frac{1}{6} \{ g_{Jjk} \bar{g}_{Jlm} g_{Klm} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K\imath m} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jlm} g_{Kml} \bar{g}_{K\zeta\eta} \delta_{i\alpha} \quad (132)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{Kmi} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} g_{Kil} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{Kli} \bar{g}_{K\zeta\eta} \} \quad (133)$$

1

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (134)$$

$$\times \frac{1}{6} \left\{ 2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} + g_{Jjk} \left(g_K g_L^\dagger \right)_{i\alpha} g_{K\zeta\eta} + \quad (135)$$

$$g_{Jjk} \left(g_K g_L^\dagger \right)_{i\alpha} \bar{g}_{K\zeta\eta} + g_{Jjk} \left(g_K g_L^\dagger \right)_{i\alpha} \bar{g}_{K\zeta\eta} + g_{Jjk} \left(g_K g_L^\dagger \right)_{i\alpha} \bar{g}_{K\zeta\eta} \} \quad (136)$$

2

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (137)$$

$$\times \frac{1}{6} \{ 2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K\imath m} \bar{g}_{K\zeta\eta} \quad (138)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{Kmi} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} g_{Kil} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{Kli} \bar{g}_{K\zeta\eta} \} \quad (139)$$

3

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (140)$$

$$\times \frac{1}{6} \{ 2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K\imath m} \bar{g}_{K\zeta\eta} \quad (141)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{Kmi} \bar{g}_{K\zeta\eta} \delta_{i\alpha} + g_{Jjk} \bar{g}_{Jl\alpha} g_{Kil} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{Kli} \bar{g}_{K\zeta\eta} \} \quad (142)$$

4

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (143)$$

$$\times \frac{1}{6} \{ 2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K\imath m} \bar{g}_{K\zeta\eta} \quad (144)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{Kmi} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} g_{Kil} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{Kli} \bar{g}_{K\zeta\eta} \} \quad (145)$$

5

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (146)$$

$$\times \frac{1}{6} \{2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K i m} \bar{g}_{K\zeta\eta} + \quad (147)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{K m i} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{J l \alpha} g_{K i l} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{J l \alpha} g_{K l i} \bar{g}_{K\zeta\eta}\} \quad (148)$$

6

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (149)$$

$$\times \frac{1}{6} \{2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K i m} \bar{g}_{K\zeta\eta} \quad (150)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{K m i} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{J l \alpha} g_{K i l} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{J l \alpha} g_{K l i} \bar{g}_{K\zeta\eta}\} \quad (151)$$

1

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k \leq r \\ 1 \leq \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (152)$$

$$\times \frac{1}{6} \left\{ 2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} + 4g_{Jjk} \left(g_K g_L^\dagger \right)_{i\alpha} g_{K\zeta\eta} \right\} \quad (153)$$

$$= \frac{1}{48} \sum_{1 \leq J \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k \leq r \\ 1 \leq \zeta, \eta \leq r}} g_{Jjk} \bar{g}_{J\zeta\eta} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_i \varphi_\zeta \varphi_\eta| \quad (154)$$

$$+ \frac{1}{24} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k \leq r \\ 1 \leq \alpha, \zeta, \eta \leq r}} g_{Jjk} \left(g_K g_J^\dagger \right)_{i\alpha} g_{K\zeta\eta} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (155)$$

2

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (156)$$

$$\times \frac{1}{6} \left\{ 2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} + g_{Jjk} \left(g_K g_J^\dagger \right)_{i\alpha} \bar{g}_{K\zeta\eta} \quad (157)$$

$$g_{Jjk} \left(g_K g_J^\dagger \right)_{i\alpha} \bar{g}_{K\zeta\eta} + g_{Jjk} \left(g_K g_J^\dagger \right)_{i\alpha} \bar{g}_{K\zeta\eta} + g_{Jjk} \left(g_K g_J^\dagger \right)_{i\alpha} \bar{g}_{K\zeta\eta} \right\} \quad (158)$$

$$= \frac{1}{48} \sum_{1 \leq J \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k \leq r \\ 1 \leq \zeta, \eta \leq r}} g_{Jjk} \bar{g}_{J\zeta\eta} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_i \varphi_\zeta \varphi_\eta| \quad (159)$$

$$+ \frac{1}{24} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k \leq r \\ 1 \leq \alpha, \zeta, \eta \leq r}} g_{Jjk} \left(g_K g_J^\dagger \right)_{i\alpha} g_{K\zeta\eta} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (160)$$

3

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (161)$$

$$\times \frac{1}{6} \{2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K i m} \bar{g}_{K\zeta\eta} \quad (162)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{K m i} \bar{g}_{K\zeta\eta} \delta_{i\alpha} + g_{Jjk} \bar{g}_{Jl\alpha} g_{K i l} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{K l i} \bar{g}_{K\zeta\eta}\} \quad (163)$$

4

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (164)$$

$$\times \frac{1}{6} \{2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K i m} \bar{g}_{K\zeta\eta} \quad (165)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{K m i} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} g_{K i l} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{K l i} \bar{g}_{K\zeta\eta}\} \quad (166)$$

5

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (167)$$

$$\times \frac{1}{6} \{2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K i m} \bar{g}_{K\zeta\eta} + \quad (168)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{K m i} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} g_{K i l} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{K l i} \bar{g}_{K\zeta\eta}\} \quad (169)$$

6

$$(A_2 \wedge P_{\mathcal{H}^1})^2 = \frac{1}{16} \sum_{1 \leq J, K \leq \binom{r}{2}} \sum_{\substack{1 \leq i, j, k, l, m \leq r \\ 1 \leq \alpha, \beta, \gamma, \zeta, \eta \leq r}} |\varphi_i \varphi_j \varphi_k\rangle \langle \varphi_\alpha \varphi_\zeta \varphi_\eta| \quad (170)$$

$$\times \frac{1}{6} \{2g_{Jjk} \delta_{JK} \bar{g}_{K\zeta\eta} \delta_{i\alpha} - g_{Jjk} \bar{g}_{J\alpha m} g_{K i m} \bar{g}_{K\zeta\eta} \quad (171)$$

$$g_{Jjk} \bar{g}_{J\alpha m} g_{K m i} \bar{g}_{K\zeta\eta} + g_{Jjk} \bar{g}_{Jl\alpha} g_{K i l} \bar{g}_{K\zeta\eta} - g_{Jjk} \bar{g}_{Jl\alpha} g_{K l i} \bar{g}_{K\zeta\eta}\} \quad (172)$$